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Work sample on Leucine's product areas · vyshnavia2k4@gmail.com · github.com/achi-vyshnavi28

In one line: I built working versions of three problems Leucine solves (batch execution with a Part 11 validation package, a spec-review tool that catches untestable requirements, and an AI root-cause analyst made auditable for GxP), then applied the same approach to one of Leucine's product areas: a spec review of a production-logbook feature (pages 2–4).

Your products, my evidence

Leucine (from leucine.io)	What I built	Proof you can check
Batch Execution	BatchGuard: an electronic batch record	21 CFR Part 11 e-signatures, hash-chained audit trail, two-person rule, release blocking. Validated like a GxP system: URS → FRS → risk assessment → IQ 17/17 · OQ 12/12 · PQ 5/5 , every requirement traced to a passing test.
Production Logbooks	BatchGuard 1.1 + this work sample	Cleaning-log PRD, spec review, UAT plan, enablement one-pager; Jira backlog built from the review; 6 Figma screens.
AI Investigator	RootCause: an AI analyst for "why did this metric change?"	50-question benchmark: 100% SQL accuracy, 93% root cause found first, 0% invented numbers . v1.2 assessed it for GxP use (GAMP 5, AI risk assessment, ALCOA+) and added a tamper-evident answer log and a model change-control gate. Live demo .
Quality · spec quality	SpecCheck: finds untestable requirements before development	Measured on 60 held-out requirements written after the rules were frozen: rules 90% precision (0 false alarms on clean requirements), rules + AI 100% recall . Includes environmental monitoring and instrument requirements.

3

working projects, public on GitHub

170+

automated tests (pytest, Playwright, Postman)

8

gaps I found in my own work, each logged with root cause

13

gaps found in a logbook spec before development (page 2)

Links

All documents (Notion)	Validation packages, PRDs, reviews, evaluations: one workspace
Code	BatchGuard · SpecCheck · RootCause
Design (Figma)	13 screens across the three products
This work sample's files	Draft and reviewed PRDs · test libraries (Excel) , traceability , Jira import

Work sample: spec review of a production-logbook feature

I wrote a first-draft PRD for equipment usage and cleaning logs, in the style a product team circulates: mostly clear, with the gaps real drafts tend to have. (My own illustration, not a Leucine document.) Then I reviewed it with three independent reviewers: SpecCheck's rules, AI-drafted test cases, and my own reading.

13 gaps found in 12 acceptance criteria	3 blocking: development should not start	12 / 12 criteria got an AI-drafted test ("100% coverage")	7 / 12 of those tests rest on an assumption
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The finding that matters most: coverage looked perfect. But for "The supervisor can edit entries as needed", the AI even noted that the edit time window was undefined, *and then wrote a boundary test for it anyway*. The test looks rigorous, yet it quietly encodes a product decision nobody made. A plain coverage % would have called this story done. The traceability matrix marks such criteria **assumed** instead of **yes**.

Blocking gaps and the questions I would send back

Story	Requirement as drafted	Why it can't be tested	Question for product / QA
LB-404	"The supervisor can edit entries as needed."	HIGH A GMP record change with no reason, e-signature or audit trail (Part 11 §11.10(e), §11.50)	Must a correction require a reason, e-signature and audit entry? Who may correct? Is the original kept?
LB-404	same	HIGH "as needed" has no pass/fail criterion	Under which exact conditions? Can entries of a released batch be corrected?
LB-403	"Clean hold time is TBD per equipment."	HIGH Not decided	Hold time per equipment type, counted from the start or the end of cleaning?

Other findings (10), and what each reviewer caught

Finding	Rules	AI	Me	Note
"Cleaning is verified." / "The entry is signed." (by whom?)	✓	✓	✓	ALCOA+ Attributable; also: can the verifier be the operator?
4 stories say nothing about what happens when a rule is broken	✓	–	✓	No negative behaviour means no negative tests
"QA reviews logbooks periodically."	✓	✓	✓	How often, and does it gate batch release?
"Recording should be quick on the shop floor."	"should" only	✓	✓	The rules missed "quick" (their list has "quickly"). Each reviewer catches different things, so I use all three
"clean hold expired" with no value	✓	✓	✓	Same root cause as the TBD

Full output: [spec review](#), [Excel test library with traceability](#), [Jira import](#).

Work sample: the reviewed spec

I rewrote the draft so every criterion can be tested, and listed the decisions only product or QA can make as open questions **with a proposed default**, instead of guessing them into the spec.

Before (LB-404)
"The supervisor can edit entries as needed."

After (LB-404)

- Only a supervisor or QA can correct an entry; an operator's attempt is refused.
- A correction requires a reason of ≥ 10 characters and an e-signature (meaning "Corrected").
- The original value stays visible; the audit trail records user, time, old value, new value, reason.
- Entries linked to a released batch cannot be corrected; the attempt is refused and QA is emailed within 15 minutes.

Before (LB-402)
"Cleaning is verified." · "The entry is signed."

After (LB-402)

- The operator signs (meaning "Performed"). A supervisor verifies a major cleaning (meaning "Verified") and cannot be the operator who performed it.
- Until a major cleaning is verified, the equipment stays "dirty". A cleaning entry without the SOP is refused, naming the missing field.

Result after the rewrite

	Draft	Reviewed
Acceptance criteria	12	20 (negative, permission and boundary behaviour now specified)
Gaps found by the rules	13 (3 blocking)	3 medium, 0 blocking: all 3 reviewed and dismissed with a reason (see below)
Criteria tested on an assumption	7 of 12	0 real: the 3 flagged are the dismissed ones
AI-drafted test cases	25	22, each linked to the criteria it covers

Not blindly following the tool: the 3 remaining flags come from rules that read one sentence at a time (the actor and the 72 h value sit in the neighbouring criterion), so I dismissed them with that reason rather than rewriting good requirements to satisfy a checker.

Open decisions (need product/QA sign-off)

Question	Proposed default	Why it matters
Clean hold time per equipment type	72 h from the end of cleaning	Tests were assuming a value the spec never set
Correct entries after batch release?	No; raise a deviation instead	Released records stay locked
Does a minor cleaning need verification?	No, only major	Changes the two-person rule and the test count
Status after a verified major cleaning	"clean", hold timer starts	<i>Raised by the AI in round two:</i> the rewrite still only said "dirty until verified"

Work sample: test cases, traceability and risk

Selected test cases (AI-drafted, then reviewed by me)

ID	Covers	Type	Steps	Expected result
TC-404-02	AC 1	permission	Log in as an operator; open an entry; try to correct it	Refused with an access-denied message; no correction screen opens
TC-404-03	AC 2	negative	As supervisor, correct an entry with the reason "Typo" (4 characters)	Validation error requiring ≥ 10 characters; nothing saved
TC-404-04	AC 4	negative	Try to correct an entry linked to a released batch	Refused; QA emailed within 15 minutes
TC-402-04	AC 2	permission	Operator X records a major cleaning, then tries to verify it	Refused: the performer cannot verify their own cleaning
TC-402-05	AC 3	boundary	Check equipment status right after a major cleaning is signed, before verification	Status is still "dirty"
TC-405-04	AC 4	permission	QA user who recorded an entry for the batch opens its logbook review	Review and sign disabled, with a message explaining segregation of duties

My review of the AI's drafts: one expected result read "equipment status updates accordingly". That can't be checked, so I sent it back as open decision 4. AI drafting saves hours, but every draft stays "draft" until a person has read it.

Traceability (excerpt)

Story	AC	Acceptance criterion	Test cases	Covered
LB-404	2	A correction requires a reason of ≥ 10 characters and an e-signature	TC-404-01, TC-404-03	yes
LB-402	2	A supervisor verifies a major cleaning; not the same person	TC-402-02, TC-402-04	yes
LB-403	3	When the hold time passes, the status changes to "clean hold expired"...	TC-403-02, 03, 05	assumed*
LB-401	5	A usage entry takes no more than 3 taps after scanning the tag	–	NO

* flag reviewed and dismissed (value defined in the neighbouring criterion). The untested criterion is honest: the AI asked what counts as a "tap", which needs the wireframe before a test can be written.

Risk assessment (GAMP 5 style, $RPN = \text{severity} \times \text{probability} \times \text{detectability}$)

Failure	Impact	RPN	Control in the reviewed spec → test
Batch started on equipment whose clean hold expired	Cross-contamination	18	Status auto-changes and usage is refused (LB-403 AC 3) → boundary test at hold time ± 1 min
Correction overwrites the original entry	Loss of original record (ALCOA+ Original)	18	Original kept + audit trail (LB-404 AC 3) → audit test
Operator verifies their own major cleaning	Independent check bypassed	12	Two-person rule (LB-402 AC 2) → permission test
Batch released without logbook review	Unreviewed records released	12	Release blocked until QA signs (LB-405 AC 1) → negative test

Gaps I found in my own work, and raised

In a validated product, a gap noticed late becomes a deviation. Here is what I noticed in my own projects, each logged with its root cause, not quietly fixed.

ID	What I found	Root cause → what changed
BatchGuard DEV-001	The ALCOA+ "Consistent" check failed on a compliant batch	Corrections overwrote the <i>observation</i> time with the correction time → original observation time kept; OQ re-run 12/12
RootCause BUG-002 S1	Every question failed, but the API still returned HTTP 200	A security-policy update blocked three libraries. Only caught because my Postman suite checks the <i>answer content</i> , not only status codes
RootCause CI	The CI pipeline had been red since a module was added, and nobody had noticed	A test dependency was missing from CI → fixed; green for the first time
RootCause BUG-003	A test passed, but its screenshot didn't show what the test claimed	In validation, evidence that doesn't show the result is a finding → test fixed
RootCause BUG-001	A button label was cut off (spotted in test evidence)	Fixed, with a regression check confirmed to fail on the old layout
SpecCheck	Two false alarms on new PRDs ("due date", "cannot be closed")	Fixed, then re-scored the held-out set to prove the blind numbers didn't change
SpecCheck	Rules scored 97% on the data they were tuned on	Wrote a held-out set <i>after</i> freezing the rules → honest score 90% / 70%, both published
RootCause GxP	4 gaps between the tool and GxP use (no user identity, no append-only permissions, no periodic review, supplier)	Stated openly in the risk assessment and planned in Jira, not hidden

What these findings suggest for a GxP logbook or AI product:

- **Report "tested on an assumption" separately from coverage.** In the logbook draft, 100% coverage hid 7 of 12 criteria whose tests encoded decisions nobody had made.
- **A correction must keep the original observation time, not just the original value.** BatchGuard's DEV-001 showed a correction silently breaking ALCOA+ "Consistent".
- **Treat the AI model as a supplier that can change.** Record the model on every AI answer and gate model changes on a benchmark (RootCause v1.2); the provider retired a model mid-project.
- **Check content, not status codes.** RootCause's API kept returning HTTP 200 while every answer failed; only a check on the answer itself caught it.

Everything above links to working code and documents. · Vyshnavi Achi · vyshnavia2k4@gmail.com